

Please complete all of the following problems.
Show all your work. Consult the textbook if you need more information.

8.6 Which of the quantum numbers relate(s) to the electron only?
Which relate(s) to the orbital?

8.9 Define *shielding* and *effective nuclear charge*. What is the connection between the two?

8.11 How many electrons in an atom can have each of the following quantum number or sublevel designations?

- (a) $n = 2, l = 1$ (b) $3d$ (c) $4s$

8.13 How many electrons in an atom can have each of the following quantum number or sublevel designations?

- (a) $4p$ (b) $n = 3, l = 1, m_l = +1$ (c) $n = 5, l = 3$

8.16 State Hund's rule in your own words, and show its application in the orbital diagram of the nitrogen atom.

8.18 For main-group elements, are outer electron configurations similar or different within a group? Within a period? Explain.

8.20 What is the electron capacity of the n th energy level? What is the capacity of the fourth energy level?

8.21 Write a full set of quantum numbers for the following:

- (a) The outermost electron in an Rb atom
(b) The electron gained when an S^- ion becomes an S^{2-} ion
(c) The electron lost when an Ag atom ionizes
(d) The electron gained when an F^- ion forms from an F atom

8.23 Write the full ground-state electron configuration for each:

- (a) Rb (b) Ge (c) Ar

8.25 Write the full ground-state electron configuration for each:

- (a) Cl (b) Si (c) Sr

8.27 Draw an orbital diagram showing valence electrons, and write the condensed ground-state electron configuration for each:

- (a) Ti (b) Cl (c) V

8.29 Draw an orbital diagram showing valence electrons, and write the condensed ground-state electron configuration for each:

- (a) Mn (b) P (c) Fe

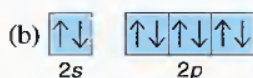
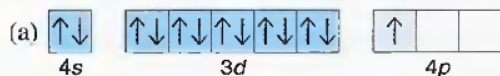
8.31 Draw the partial (valence-level) orbital diagram, and write the symbol, group number, and period number of the element:

- (a) $[\text{He}] 2s^2 2p^4$ (b) $[\text{Ne}] 3s^2 3p^3$

8.33 Draw the partial (valence-level) orbital diagram, and write the symbol, group number, and period number of the element:

- (a) $[\text{Ne}] 3s^2 3p^5$ (b) $[\text{Ar}] 4s^2 3d^{10} 4p^3$

8.35 From each partial (valence-level) orbital diagram, write the ground-state electron configuration and group number:



8.37 How many inner, outer, and valence electrons are present in an atom of each of the following elements?

- (a) O (b) Sn (c) Ca (d) Fe (e) Se

8.39 Identify each element below, and give the symbols of the other elements in its group:

- (a) $[\text{He}] 2s^2 2p^1$ (b) $[\text{Ne}] 3s^2 3p^4$ (c) $[\text{Xe}] 6s^2 5d^1$

8.46 Explain the relationship between the trends in atomic size and in ionization energy within the main groups.

8.48 Why do successive IEs of a given element always increase? When the difference between successive IEs of a given element is exceptionally large (for example, between IE_1 and IE_2 of K), what do we learn about its electron configuration?

8.53 Arrange each set in order of *increasing* atomic size:

- (a) Rb, K, Cs (b) C, O, Be (c) Cl, K, S (d) Mg, K, Ca

8.55 Arrange each set of atoms in order of *increasing* IE_1 :

- (a) Sr, Ca, Ba (b) N, B, Ne (c) Br, Rb, Se (d) As, Sb, Sn

8.57 Write the full electron configuration of the Period 2 element with the following successive IEs (in kJ/mol):

$$\text{IE}_1 = 801 \quad \text{IE}_2 = 2427 \quad \text{IE}_3 = 3659$$

$$\text{IE}_4 = 25,022 \quad \text{IE}_5 = 32,822$$

8.61 List three ways in which metals and nonmetals differ.

8.62 Summarize the trend in metallic character as a function of position in the periodic table. Is it the same as the trend in atomic size? Ionization energy?